

KHRUL SERGEI



TECHNICAL LEAD & RESEARCHER

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ABOUT

Hi! I am a visionary engineering leader and hands-on CTO with 18+ years of experience delivering **high-load**, scalable, and **secure** systems across **Web3**, **FinTech**, and **iGaming**. Proven track record of bridging **business objectives** with technical execution: from driving **R&D** and securing strategic partnerships (Consensys, MetaMask), to rescuing failing projects through crisis management. Adept at scaling engineering teams (from 4 to 25+ FTEs), optimizing IT infrastructure, and **architecting** distributed systems capable of handling 10,000+ concurrent sessions and millions in trading volume.

SKILLS & OTHER

- **Engineering Leadership:** Technical Strategy & Roadmapping, Crisis Management, Team Building & Mentoring, Product sEngineering, Vendor Management, Agile/Scrum.
- **Architecture:** Microservices, High-Load & HFT Systems, Event-Driven Architecture, Smart Contracts (Web3/DeFi), Data Engineering.
- **Core Tech Stack:** Java (11-25), GoLang, TypeScript, C#, Solidity, React, Angular, React Native.
- **Data & Messaging:** PostgreSQL, ClickHouse, MongoDB, Redis, Kafka, RedPanda, AWS SQS, HazelCast.
- **DevOps & Infrastructure:** AWS (EKS, EC2, Lambda) + Terraform, Kubernetes, Docker, Jenkins, Grafana, Prometheus.
- **Frameworks:** Spring, Protobuf, Reactor, Swagger, Junit, Unigine, WPF, Selenium.

PROFESSIONAL EXPERIENCE

Tech Lead, Freelance / B2B (EU)

2025 - present

Main projects: High-load MMO football game with WebGL support.

- *Team Scaling & Management:* Built and mentored a cross-functional R&D department from scratch, scaling the engineering team from 4 to 18 specialists (Backend, Frontend, QA, DevOps).
- *Business-to-Tech Alignment:* Acted as the crucial bridge between product stakeholders and engineering, translating business requirements (Figma, BRDs) into robust technical architectures and successfully executing 5 major product epics.
- *High-Load Architecture:* Rescued a legacy monolith by migrating to Java 25 (Virtual Threads) and Hazelcast 5.6. Containerized the infrastructure to Kubernetes, optimizing throughput from 30 to **4,000 RPS** per instance and enabling **10,000+ concurrent** user sessions with near-zero downtime on deploy.

Acting CTO / Crisis Manager, Gydde - Web3 Startup (EU)

2024 – 2025

Main projects: 1) Gamified meme-token launchpad with ERC-4337 AO wallet integration; 2) Crypto Educational Platform for Telegram Mini-App.

- *Crisis Management & Recovery:* Successfully led platform rescue after a critical infrastructure and team loss. Coordinated a task force to reverse-engineer and decompile 5,000+ backend files from legacy Docker images, preventing data loss and migrating the platform to AWS within 4 days.
- *Product Innovation (R&D):* Spearheaded the rapid prototyping of a Telegram Mini App with an integrated wallet and biometric-secured key storage. Managed the delivery of the Memetoken Launchpad to production within a strict 2-month timeline.
- *Technical Strategy & Business Growth:* Formulated and executed the company's long-term technical vision and development strategy. This strategic alignment between engineering and business goals formed the core of the investment memorandum, ultimately securing the company's place in the **Consensys Scale Program** and achieving **MetaMask Alpha Partner** status.
- *Anti-Cheat ML System:* Engineered an ML-driven fraud detection and anti-bot system, resulting in over 600 daily user bans.

Tech Lead, Firmshift B.V (EU)

2022 - 2025

Main projects: 1) Omnichain protocol on Cosmos SDK; 2) Analytical platform for multichain DEX; 3) HFT Trading bot for CEX-DEX arbitrage.

- *Omnichain Protocol* (TVL \$12M+): Led the technical recovery and launch of a stalled cross-chain bridge on Cosmos SDK across 7+ networks with gas abstraction. Engineered core optimizations that reduced transaction costs to \$0.30 and execution time to <10 seconds, driving strategic partnerships with the **Berachain** and Polygon Foundations.
- *High-Performance Analytics:* As a solo developer, built a multichain analytical platform with a custom cross-chain indexer (5 networks, latency <30ms) from scratch in just 3 months. Integrated full support for **TheGraph**'s GraphQL format, enabling seamless migration for partners from external indexers.
- *HFT Trading Core Development:* Acted as Sole Architect to build an in-house AMM and CEX-DEX arbitrage platform, eliminating costly vendor lock-in. The system efficiently processed **\$200k in daily trading volume**.
- *Risk Management & MEV:* Designed a Risk Engine with automated circuit breakers for high market volatility and implemented MEV-protected RPC integrations with dynamic slippage algorithms.

Technical Manager / Lead Full Stack Developer, Leroy Merlin / Bergmann InfoTech (FR/DE)

2017-2020

Main projects: 1) Cyber-sports tournament arena with crypto accounting; 2) SaaS platform for digital factoring; 3) National-scale personal data management system; 4) Web3 NFT-marketplace; 5) Mobile e-commerce platform (React Native).

- *Full-Stack & Mobile Development:* Hands-on architect and lead developer for web portals (React, Angular) and cross-platform mobile apps (React Native), ensuring seamless integration with microservice backends.
- *Project Leadership:* Guided the end-to-end development of 10+ enterprise digital platforms (ranging from video analytics to FinTech factoring applications).
- *Talent Acquisition:* Revolutionized the hiring pipeline, successfully recruiting over 20 developers and managing a cohesive, high-performing team of 25 engineers.
- *Performance Optimization:* Achieved a 60% increase in big-data search speed across heterogeneous sources and modernized CI/CD workflows for large-scale distributed systems.

Software Engineer, Team Lead, HUGHES Satellite Monitoring Center (US, RU)

2010 - 2017

Main projects: 1) Real-time public transport monitoring. 2) R&D for ARM-based HW and SW for autonomous vehicles.

- *R&D Leadership & Grants:* Led advanced R&D projects in robotics and telecommunications, successfully winning and managing grants exceeding **\$10M+**.
- *Innovation & Accuracy:* Boosted GPS tracking accuracy by 50% using custom inertial-based algorithms and secured 3 software patents for autonomous vehicle technologies.
- *Team Building & Mentorship:* Successfully delivered a cross-platform taxi-booking frontend (Java/Swift) by strategically distributing tasks among 3 part-time interns, ultimately securing full-time offers for the entire team.

Software Engineer Intern, Content Manager, Private Companies (RU)

2003 – 2004, 2008 - 2010

Main projects: 1) Turnstile-based access control systems. 2) Video streaming and web storage platform.

- *System Maintenance & Optimization:* Maintained and optimized a turnstile access system (C++, SQL, PERCo devices).
- *Feature Development & Monetization:* Achieved a 30% increase in monthly revenue by modifying and integrating new media content features into a web portal.

EDUCATION

- **Tomsk Polytechnic University** (RU), Post-Graduate in Computer Engineering 2014
Degree topic: Satellite monitoring system for autonomous transport
- **Tomsk State University** (RU), Master's degree in Applied Mathematics. 2010

ACHIEVEMENTS & CERTIFICATIONS

- Won personal state R&D grant for 2 years
- 17 publications in Europe and Russian journals
- Blockchain Fundamentals by ChainLink
- Advanced Solidity by cryptozombies.io
- Artificial Intelligence for Robotics by Udacity
- Solidity Smart Contract Guide by BitDegree
- Learn Solidity with Space Doggo by BitDegree
- Self-Driving Car Engineer Nanodegree by Udacity